

Sample Data Document

Project Title

IncidentIQ – AI Powered Incident Intelligence & Root Cause Analysis Generator

Purpose of Sample Data

The following sample datasets were used to validate the RCA generation workflow and demonstrate the system's ability to identify root causes from incident timelines, error logs, and code changes.

Sample Dataset 1

Incident Timeline

10:00 AM – New deployment started

10:05 AM – Error rate increased

10:10 AM – Service unavailable

10:20 AM – Rollback initiated

10:35 AM – Service restored

Error Logs

ERROR Database connection timeout

ERROR Connection pool exhausted

ERROR HTTP 500 Internal Server Error

Git Diff

- maxPoolSize = 50
- maxPoolSize = 10
- timeout = 30s
- timeout = 5s

Expected Output

Root Cause

Database connection pool configuration was incorrectly modified during deployment.

Severity

Critical

Impact

Service unavailable for approximately 25 minutes.

Recommended Fixes

- Restore maxPoolSize configuration.
- Increase timeout values.
- Validate deployment configuration before release.

Prevention Actions

- Perform load testing before deployment.
 - Introduce configuration review procedures.
 - Enable automated rollback mechanisms.
 - Improve monitoring and alerting systems.
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Sample Dataset 2

Incident Timeline

09:00 AM – Release deployed

09:10 AM – API latency increased

09:20 AM – Multiple timeout errors observed

09:30 AM – Rollback initiated

09:45 AM – Service recovered

Error Logs

API Gateway Timeout

Redis Cache Miss

Application Timeout Error

Git Diff

- cacheTTL = 300
- cacheTTL = 10
- requestTimeout = 30s
- requestTimeout = 3s

Expected Output

Root Cause

Configuration changes caused excessive cache invalidation and request timeout failures, resulting in degraded application performance.

Severity

High

Impact

Increased latency and temporary service degradation.

Recommended Fixes

- Restore cache configuration values.
- Increase request timeout limits.
- Validate configuration changes through testing.

Prevention Actions

- Conduct performance testing.
- Introduce staged deployments.
- Implement automated configuration validation.